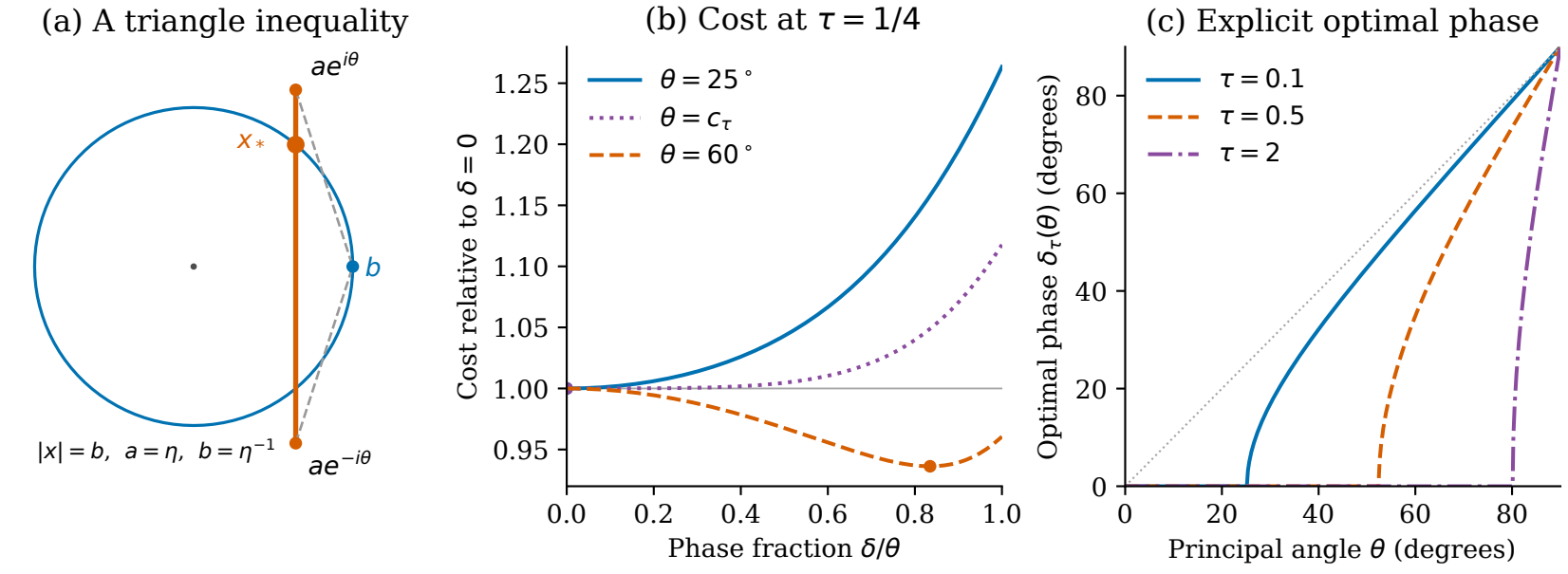
(a) A triangle inequality
$ae^{i\theta}$
x_*
b
$|x| = b,\ a = \eta,\ b = \eta^{-1}$
$ae^{-i\theta}$
(b) Cost at $\tau = 1/4$
$\theta = 25^\circ$
$\theta = c_\tau$
$\theta = 60^\circ$
Cost relative to $\delta = 0$
1.25
1.20
1.15
1.10
1.05
1.00
0.95
0.0
0.2
0.4
0.6
0.8
1.0
Phase fraction δ/θ
(c) Explicit optimal phase
$\tau = 0.1$
$\tau = 0.5$
$\tau = 2$
Optimal phase $\delta_\tau(\theta)$ (degrees)
80
60
40
20
0
0
20
40
60
80
Principal angle θ (degrees)